



OneATS

User manual

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株式会社 東光高岳
TAKAOKA TOKO CO., LTD.



OneATS user manual

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Definitions, symbols, abbreviations

Symbol	Description
CIM	Common Information Model
DMS	Distributed Management System
DCOM	Distributed Component Object Model
FEP	Front End Processing
HMI	Human Machine Interface
I/O	Input/Output
IED	Intelligent Electronic Device
LAN	Local Area Network
OPC UA	OPC Unified Architecture
RTU	Remote Terminal Unit
SCADA	Supervisory Control and Data Acquisition
SSL	Secure Socker Layer
TCP	Transmission Control Protocol
TLS	Transport Layer Security
XML	Extensible Markup Language

Text color:

Comment

Notes

Warnings

Chapter	Content
A.	System overview
B.	Database
C.	OneATS DataEditor – Project Management
D.	OneATS FEP
E.	OneATS DataServer
F.	OneATS Grid Studio
G.	OneATS SmarHIS
H.	OneMasterTool



OneATS User Manual

- SYSTEM OVERVIEW

12/Jun/2024

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Software version	Date of Issue	Change Description
V4.2	Q4-2023	Add more new functions, new system features for various requirements



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A. SYSTEM OVERVIEW

OneATS Platform is **a unified and comprehensive platform** for monitoring, operation and management of distribution power grids. OneATS Platform is researched and developed by ATS, drawing from decades of experience in implementing projects from stations to regional and regional control centers.

OneATS Platform leverages the most modern technologies, combined with years of experience in researching, developing and deploying systems on all scales of ATS, to deliver the most satisfying experiences to customers through added value.

A.1. General Description

Standardized system

OneATS Platform uses international standards such as IEC-62541 (OPC UA), IEC-60870-5-101/104, CIM (Common Information Model), TLS (Transport Layer Security) ,... to ensure security, reliability, compatibility and scalability for the system. The system fully meets the standards and policies set by the NERC-CIP/IEC 62351 council on system security and safety.

High performance

OneATS Platform was designed and built entirely from the beginning by ATS with many years of experience in researching and developing products of all levels and scales. The most advanced techniques are applied to the system such as time-series specialized database, NoSQL database, In-memory Caching, parallel processing, ensure high-performance processing capability for the whole system.

High availability

With redundancy across all operating servers, OneATS Platform provides a system that meets high availability, ensuring the system works 24/7.

Fast-growing platform

OneATS Platform provides users with a variety of project building tools quickly. With tools such as Object Templates, configuring a place to use everywhere, mechanisms for detecting and correcting milk configuration errors, run-time debugger, fast cloning, Project development techniques will significantly improve productivity.

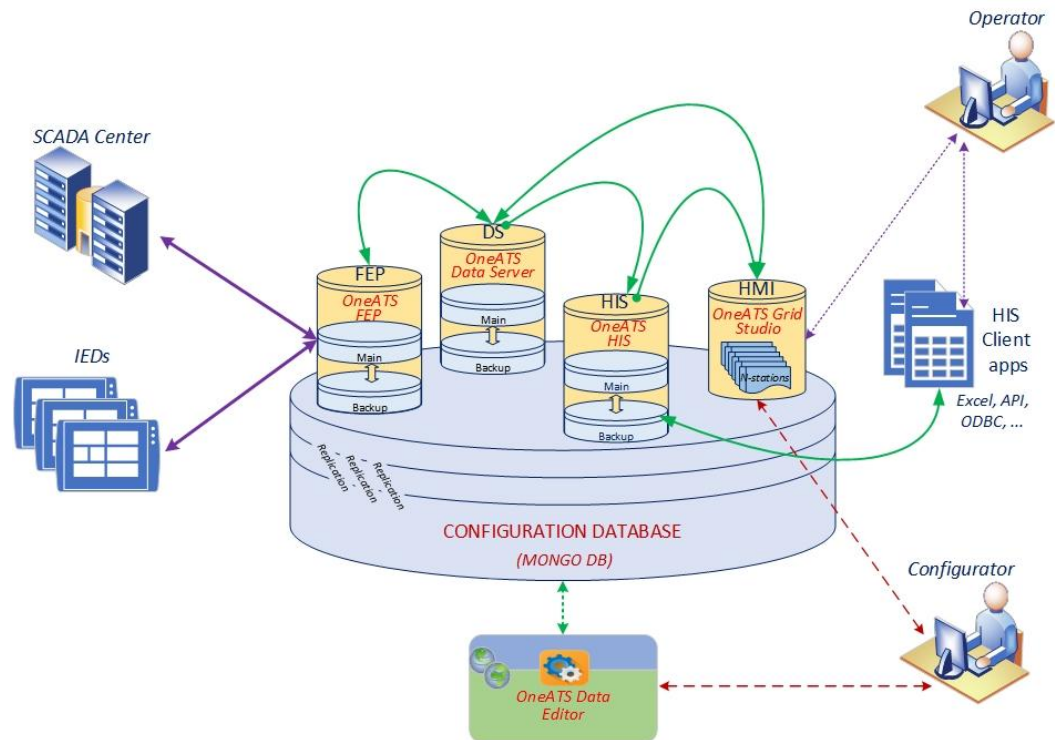
High customizable and extensible

OneATS Platform is capable of meeting diverse needs for every type of project from station to control center on all scales. Diverse customization capabilities in the process of configuring data, events, alerts, objects, will make the system flexible and easy to meet the current and future needs of the electrical system.

A.2. System structure

OneATS Platform is modular-designed, operating under an event-driven system, based on a unified platform.

The following diagram describes the main components in the OneATS Platform



FigureA-1 OneATS System Structure

Component modules in the system

Configuration Database	The configuration database of the project (data configuring FEP channels, DataModel data model configuration data, configuration of modules). The configuration of modules on a database ensures uniformity across all modules that exploit the common database. The database is built on the MongoDB open source database system.
OneATS FEP	The module collects data exchange with IEDs, control centers, ... through communication protocol standards.
OneATS Data Server	Module tính toán, xử lý dữ liệu thời gian thực: ▪ Data server (point-value/quality/timestamp, method) ▪ Alarm Server ▪ Tagging Server
OneATS Grid Studio	HMI interface module.
OneATS SmartHIS Server	Past database module ▪ Historical data (point-value/quality/timestamp) ▪ Alarm/Event Message
OneATS SmartHIS Client Apps	Past data mining applications, including OneATS or third-party applications, are integrated with OneATS libraries (Excel Add-in, ODBC Interface, API, ...) to mine HIS Server data. Libraries include: ▪ (Excel add-in) ATS Datalink: is a built-in add-in on MS Excel to retrieve data from OneATS HIS Server ▪ ODBC Driver ▪ Web Service (API) ▪ Report Viewer: Application to view report
OneATS Data Editor	Module configuring and monitoring data of system components: ▪ OneATS FEP ▪ OneATS Data Server
OneATS SmartHIS Administrator Tool	Module configuring and monitoring data of past system components: ▪ OneATS SmartHIS Server

List of Services Process/Applications

Module	Program Name (in Control Panel / Programs and Features)	Program Type & Name (in Taskbar / Service)	Process Name (in Task Manager)
Configuration Database	MongoDB ...	(Service) MongoDB Server	MongoDB Database Server
OneATS FEP	OneATS Fep	(Service) OAFEP	OneATS FEP Process Manager
OneATS Data Server	OneATS DataServer	(Service) ODataServer/ OneATS Data Server Process Management Service	ODataServer.exe
OneATS Grid Studio	OneATS SmartHMI	(Desktop Application)  OneATS Grid Designer	Grid Designer
		(Desktop Application)  OneATS Grid Viewer	Grid Viewer
OneATS HIS	ATS SmartHIS	(Service) OAHISServer	ATS Smart His Server
		(Desktop Application) SmartHIS Administrator	ATS Smart HIS Administrator
HIS Client Apps	Excel – Add in	(COM Add-in) ATS DataLink UI (Excel Add-in) ATS.DataLink.Funtion Add-In	Microsoft Excel
	ODBC Data Source	(System DSN) ATS.HIS	
	IIS – Service	(Application Pools & Site) Smart HIS Webservice	Internet Information Service
	Report Viewer	(Desktop Application) 	ATS Report Viewer



		ATS Report Viewer	
OneATS Data Editor	OneATS DataEditor	(Desktop Application)  OneATS DataEditor	OneATS DataEditor

A.3. Internal Protocol - OPC UA

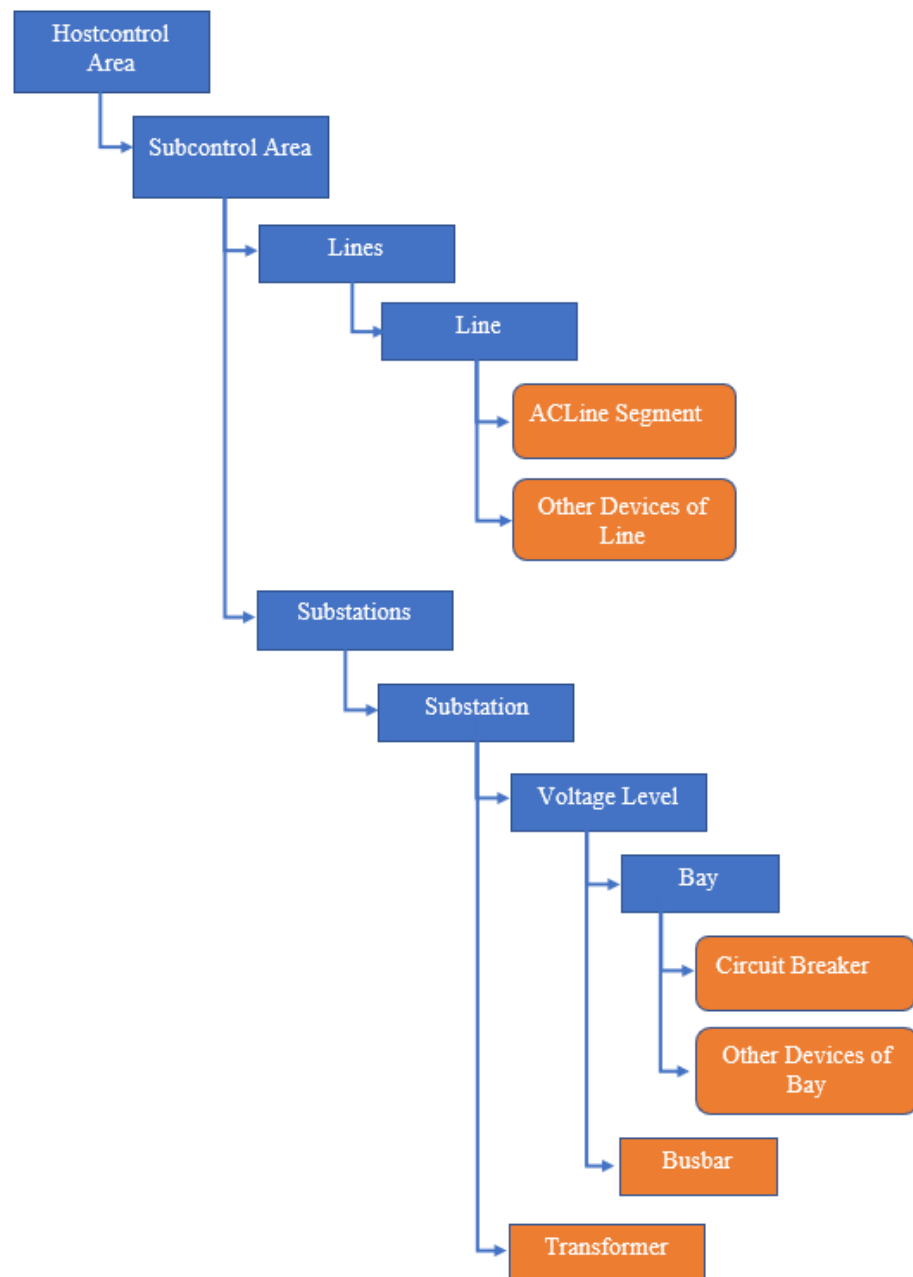
Components in the OneATS (OneATS FEP – OneATS Data Server – OneATS Grid Studio Viewer – SmartHIS) system exchange operational data (realtime data, historical data, alarm/event, program) according to OPC-UA standards.

OPC Unified Architecture (OPC UA) is a standard that ensures open connectivity, interoperability, security and reliability of industrial automation devices and systems. It works with many software platforms, is completely scalable and has been widely applied across many industries because of its versatility.

The construction of information models with OPC UA is also very flexible with object-oriented model characteristics, which are easy to scale to describe an object from simple to complex. In addition to the properties used to describe an object, another strength in opc ua's information model is the provision of methods. The method is executed on UA Servers and is called from UA Clients. According to the specification of the standard, the information model built into OPC UA includes: DA (Data Access), AC (Alarm & Conditions), HA (Historical Access), Prog (Programs).

The OneATS system uses opc UA standards to improve server-client interaction, take advantage of login control and security, support the SSL/TLS channel encryption protocol of OPC UA standard, integrate the "Alert" module into the Data Server and conveniently allow users to build data models (MODELS).

A.4. DataModel



FigureA-2.Data model structure by CIM

DataModel is a structured dataset that models electrical system data including: master bar diagram, quantity/type/order of equipment arrangement, electrical engineering connections between devices, etc.

DataModel follows the **CIM** data model (IEC 61968, 61970), which manages data in the electrical system in an object-oriented fashion.

DataModel's structured data is stored in the Database.

DataModel's realtime data is calculated and processed in the **OneATS DataServer** app.

DataModel's past data is stored in the **OneATS HIS Server** app.

A.5. Electrical Network objects

Different from the basic graphic object, graphic diagram objects are closely linked to each other in terms of operating state. Graphic Diagram objects are used to represent the operating state of systems in the form of networks, such as electrical systems, water systems, gas systems, etc. in which, the operating state of the two adjacent elements has a reciprocal relationship with each other. The link between graphic diagram elements is established and expressed by the graphics (drawing)

OneATS supports graphic diagram objects in the field of grid with the group of objects (Electrical) Network. The Electrical Network objects are closely linked to each other in terms of electrical relations (voltage, current - Kichoff Law, power ...).

The Electrical Objects (Electrical) Network is used for network topology (DMS, DAS, ...), coloring, etc.

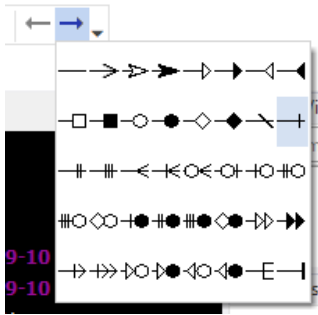
A.5.1. Electrical Network objects

OneATS hõ supports Electrical Network

No.	Object	Symbol	Describe
1.	Connectivity Node		Electrical junctions of electrical equipment, or power consumption loads, or supplies Graphic form: point
2.	Busbar		Busbar, the property on the grid is a Connectivity Node. Graphic form: straight line
3.	AC Line		Section of line
4.	Circuit Breaker	Symbol Template designed by the user	Circuit Breaker
5.	Disconnecter		Disconnectors switch
6.	Ground Disconnecter		Earth switch
7.	Transformer		Transformer (2 coils, 3 coils)
8.	Series Compensator		Series Compensator
9.	Recloser		Recloser

10.	Load Break Switch		LBS
11.	LBFCO		Cutout Fuse
12.	FCO		Fuse
13.	Energy Consumer		Load
14.	Shunt Compensator		Shunt Compensator
15.	Surge Protector		Surge protector
16.	Generator		Generator
17.	Energy Source		Energy Source
18.	Substation		Substaion
19.	Fault Indicator		Fault Indicator

In addition, OneATS supports special graphic objects, only makes sense in terms of graphical display on the grid:

No.	Object	Describe
1.	Control Arrow	There is always the same color as the Connectivity Node that it connects to. Graphic form: sucking point 

A.5.1.1 Connectivity Node.

Connectivity Node is the electrical junction of electrical devices.

Đối tượng Connectivity Node trong thực tế là: busbar, match points, the junction between lines and circuit breaker, the junction between circuit breaker and disconnectors switch, ...



Connectivity Node: it also represents a load point on the grid.

A.5.1.2 Busbar.

Busbar: as a conductor, busbar allows multiple devices with the same voltage level to be joined together.

Busbar objects are in fact the lead bar systems, busbar.



In terms of electrical properties, Busbar is a Connectivity Node.

A.5.1.3 Source.

Source is the power supply point for the grid.

The main power points in practice are generators, the starting point of the line connecting to the grid, Shunt Compensator...

In the coloring problem, The Source power point (voltage) is calculated as a landmark to determine the state of carrying electricity of elements in the grid.

Usually, based on the voltage measured at the ends of the line, the female bar, the generator pole head, ... to determine the status of elements in the grid.

In terms of electrical properties, The Source is a special Connectivity Node with a voltage state.

A.5.1.4 The concept of Connection point (Terminal)

Connection Point is the property of the device (AC Line, Circuit Breaker, Disconnecter, Earth Switch, Transformer).

Connection Point is the connection point of the device to the outside. Different types of devices will build templates with different Connection point numbers.

For example, circuit breaker, disconnecter switch is a device modeled by 2 Connection point, while Earth Switch is built with 1 Connection point (the other end is understood to be grounding so no connection point is required). The transformer can be modeled in the template with 2 or 3 Connection points depending on the number of coils (voltage levels) of the MBA.



FigureA-3. Illustration of a Connection Point object

A.5.1.5 Main electrical appliances

Object	Status data	Number of terminals on single-line	Description/ properties (laws) of voltage level state
Circuit Breaker, Disconnecter	Close-cut status	2	Terminals have the same voltage level.

			In a quality good/manual closed state, 2 termials will have the same power carrier state.
Earthing Switch	Close-cut status	1	In a closed state & good data (quality good / manual), termial has an electrical carrier state = "grounding" In the bad data state, the terminal has an electrical carrier state = "The grounding status cannot be determined"
3-windings transformer		3	Other terminals with voltage levels The termials will have the same electric carrying state
2-windings transformer		2	Other terminals with voltage levels The termials will have the same electric carrying state
AC Line		2	Terminals have the same voltage level. The termials will have the same electric carrying state

A circuit breaker extend that can be modeled as a cutting machine, in which the closing state is the synthesis of the circuit breaker state and the operating position (in/out).

A.5.2. Principles of connecting Electrical Network objects

Electrical devices (Circuit Breaker, Disconnecter, Earthing Switch, Transformer, AC Line) represented by Connection Points (Terminals) associated only with Connectivity Nodes (normal Connectivity Node, Busbar, Source)

Connection Point (Terminal) always has the same electrical states as the Connectivity Node with which it links.

Ensuring visibility, (topology) link between Connection Point (Terminal) and Connectivity Node is declared, expressed in the OneATS system by drawing technique with OneATS Grid Studio module.

A.6. System requirements

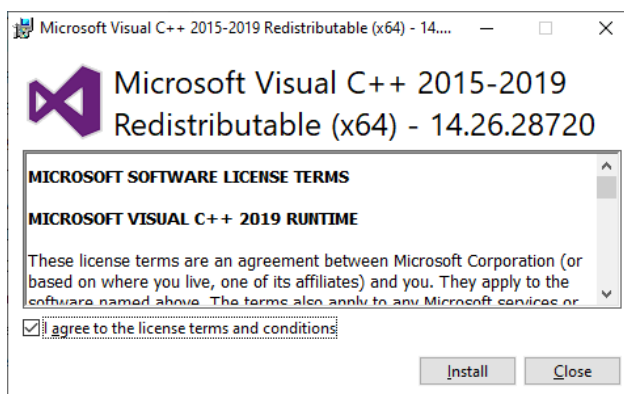
Operating system requirements & hardware capacity

Directory	Minimum	Recommendations
Server	- Windows server 2008 - Intel Xeon E4 - Memory (RAM) 8GB - HDD storage 200GB - Networking 10/100Mbps	- Windows server 2012 - Intel Xeon E-2100 series - Memory (RAM) 32GB DDR4 512GB SSD + 2TB HDD - Networking Gigabit, dual interface
Client	- Windows 7 - Intel Core i5 - Memory (RAM) 8GB - HDD 200GB - Networking 10/100Mbps	- Windows 10 + - Intel Core i7 - Memory (RAM) 32GB DDR4 512GB SSD - Networking Gigabit, dual interface

Library requirements, features that come with the operating system

Directory	Minimum
MS Visual C++ Runtime Library	<i>Microsoft Visual C++ 2015-2019 Redistributable</i>

In case the computer does not have a *Microsoft Visual C++ library 2015-2019 Redistributable*, install the library



FigureA-4 Install the MS Visual C++ Runtime Library

A.7. Security

The modules of the OneATS platform operate on the OPC UA protocol platform and industrial communication protocols (between FEP and IEDs) such as IEC61850, IEC60870-5, Modbus... and each of these protocols requires the servers to allow permission. some ports TCP, UDP bypass the firewall. Therefore, when configuring the software suite in the OneATS platform, you will have to configure this information for the system to work properly.

A.7.1. Firewall configuration

The software in the OneATS platform uses the internal protocol OPC UA. In addition, the FEP Server module itself will connect to IEDs through industrial communication protocols such as IEC61850, IEC60870-5, Modbus... These protocols use TCP/IP protocol. Therefore, it is necessary to configure the opening of specific TCP or UDP ports for these cases.

The project configurator will then perform the enable Rule configuration operations in the firewall to open the port. There will be 2 types of TCP ports that need to be configured to open ports.

A.7.1.1 OPC UA Port

- Database system: **27017** (default)
- DataServer: **48050** (default)
- HIS: **48010** (default)
- FEP: **46xxx**. Whenever creating a connection channel on FEP, FEP Server will generate 1 corresponding UA port for each channel in ascending order.

A.7.1.2 Communication Protocols Port (FEP to IED)

FEP Server connects to IEDs according to industry communication protocols. TCP ports are used as defined by the standard. The configurator based on the ports used to configure the firewall

- IEC61850: 102
- IEC60870-5-104: 2404 (default)
- ModbusTCP: 502
- ...

A.7.1.3 Firewall configuration

Open Window Defender Firewall with Advanced Security/Inbound Rules. => Select New Rules

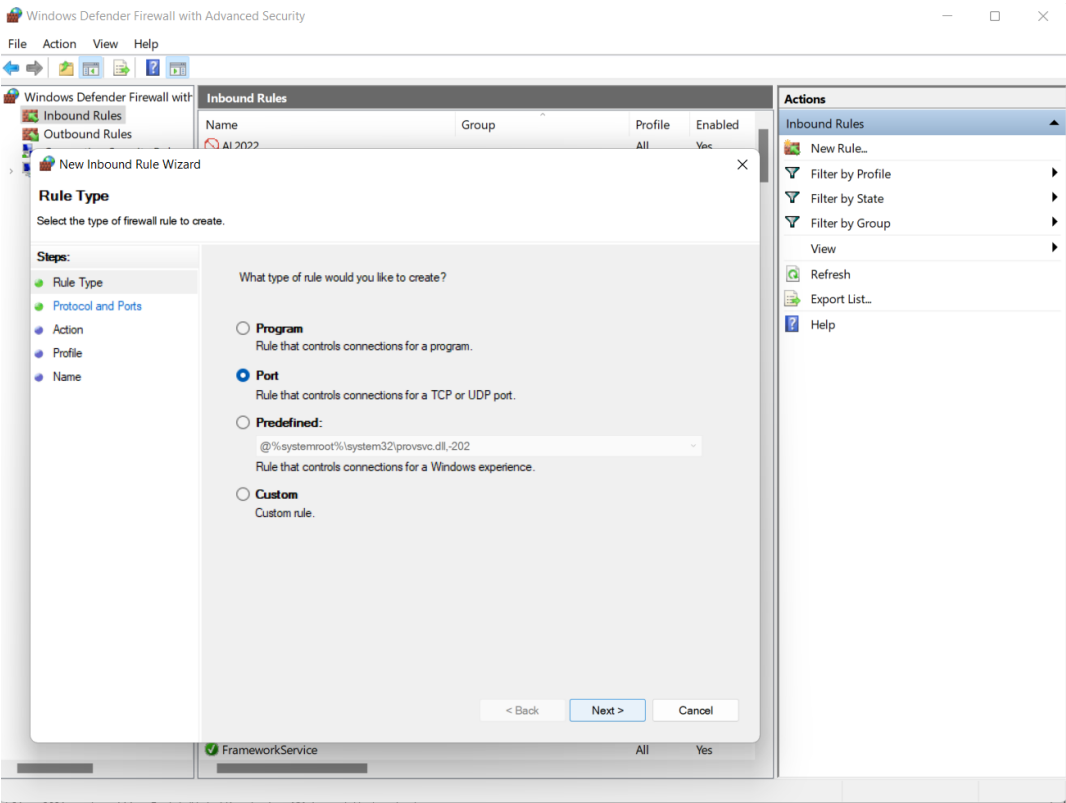


Figure A-5 Create new rule port TCP – Firewall

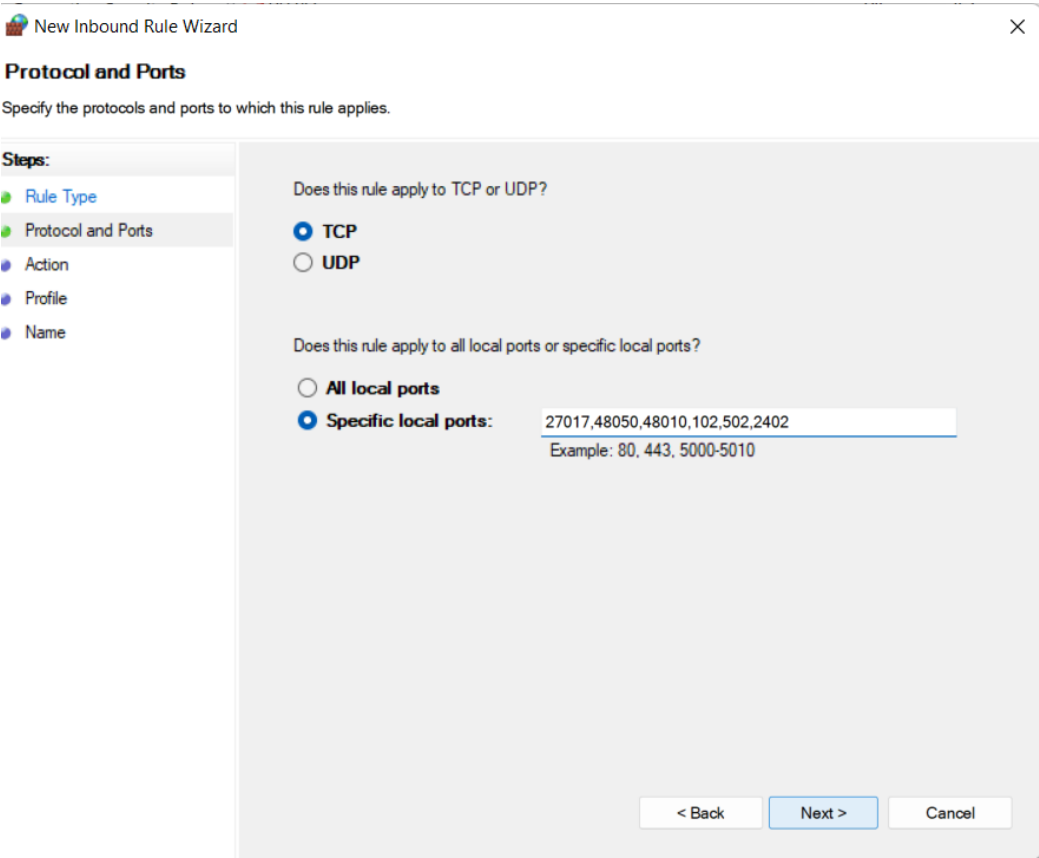


Figure A-6 Set up rule TCP Port

Choose “Allow the connection” and then name the rule. After that “Finish” to complete.



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